SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions
COURSE CODE: CSA1516

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Total Marks: 25

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Code of Conduct

I NIKITHA .R (192524021) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

1. Cloud Auto-Scaling and Load Balancing During Traffic Spikes

Introduction

E-commerce companies often experience unpredictable traffic during seasonal sales, festival offers, Black Friday events, flash sales, and other promotional campaigns. A sudden increase in users can overload application servers if the infrastructure has a fixed capacity. Cloud auto-scaling and load balancing solve this problem by dynamically adding resources and distributing requests across healthy servers.

Cloud Auto-Scaling

Auto-scaling is a cloud capability that automatically adjusts the number of computing resources according to workload demand. During normal periods, the application can run with a smaller number of instances. When traffic, CPU utilization, memory usage, or request count increases beyond a defined threshold, additional instances can be launched automatically.

- Scales out by adding additional application instances when demand increases.
- Scales in by removing unnecessary instances when demand decreases.
- Uses policies based on CPU utilization, memory, request count, queue length, or other metrics.
- Prevents the application from depending on a fixed number of servers.
- Reduces cloud expenditure because resources can be released after peak demand.

Load Balancing

A load balancer acts as an entry point for incoming application requests and distributes those requests among multiple backend servers. It can use methods such as round-robin, least connections, or health-based routing. If a server becomes unhealthy, the load balancer can stop sending requests to it.

A typical architecture is: Customers → Load Balancer → Multiple Application Servers → Database and Other Services.

How Auto-Scaling and Load Balancing Work Together

1. Users generate a large number of requests during a sale.
2. The load balancer receives and distributes those requests among available servers.
3. Monitoring detects increasing workload or utilization.
4. The auto-scaling policy launches additional application instances.
5. The load balancer adds healthy new instances to the traffic pool.
6. When the sale ends and traffic falls, auto-scaling terminates unnecessary instances.
7. The load balancer continues routing traffic only to healthy instances.

Reduction in Downtime

This approach reduces downtime by preventing individual servers from becoming a single point of failure. Traffic is distributed across multiple instances, so the failure of one instance does not necessarily make the entire application unavailable. Health checks allow unhealthy instances to be removed from service, while auto-scaling can create replacement capacity.

Improvement in User Experience

- Faster response times because workload is distributed.
- Fewer timeout and server-overload errors.
- Better availability during peak traffic.

- More reliable checkout and payment operations.
- Capacity can increase automatically without users waiting for administrators to add servers.

Role of Monitoring Tools

Monitoring is the feedback mechanism that tells the cloud platform what is happening in the application. Monitoring systems collect metrics and logs such as CPU usage, memory usage, request rate, latency, error rate, server health, network traffic, and database performance.

- Detect abnormal increases in traffic.
- Provide metrics used by auto-scaling policies.
- Generate alerts when performance or availability thresholds are exceeded.
- Help administrators identify bottlenecks and failed instances.
- Support capacity planning and post-incident analysis.

Real-World E-Commerce Scenario

Consider a large e-commerce platform such as Amazon during a major festival sale. At the start of the sale, millions of users may simultaneously browse products, search for items, add products to carts, and place orders. A fixed server environment could become overloaded. With cloud auto-scaling, additional application instances can be created as demand increases. A load balancer distributes customers across these instances, while monitoring tools track traffic, latency, errors, and resource utilization. After the sale, the platform can scale back down.

Conclusion

Cloud auto-scaling and load balancing provide an efficient way to handle unpredictable traffic. They improve availability, reduce downtime, maintain performance, and control infrastructure costs. Monitoring tools complete the solution by continuously observing system health and providing the information needed for automated scaling and operational decisions.

2. **An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.**

2. Cloud Storage, Durability, Redundancy, Security and Disaster Recovery

Introduction

Organizations generate large amounts of backup data from databases, applications, documents, customer records, configurations, and transaction systems. Cloud storage provides scalable capacity while reducing the need to maintain large physical storage infrastructures.

Durability and Data Redundancy

Durability refers to the ability of a storage service to preserve data without loss over time. Cloud providers achieve high durability through redundant storage systems, multiple copies, error detection, integrity checking, and, depending on the selected service, replication across availability zones or geographic regions.

- Multiple physical storage devices can hold redundant copies or encoded data.
- Failure of a disk does not necessarily cause data loss.
- Data may be replicated across separate availability zones.
- Critical data can be copied to another geographic region for disaster recovery.
- Automated integrity mechanisms can detect and repair certain storage failures.

Encryption of Backup Data

Encryption protects backup information from unauthorized disclosure. Encryption at rest protects data while it is stored in cloud storage, while encryption in transit protects information while it moves between systems. Organizations can also manage encryption keys using cloud key-management services and apply strict permissions to those keys.

Access Control

Access control ensures that only authorized users, applications, or administrators can access backup information. Identity and Access Management (IAM), role-based access control, least-privilege policies, multi-factor authentication, and separate administrative roles can reduce unauthorized access.

- Backup administrators can be allowed to create and restore backups.
- Regular users can be prevented from accessing backup storage.
- Delete permissions can be restricted to authorized personnel.
- Multi-factor authentication can protect privileged accounts.
- Audit logs can record access and administrative changes.

Cost Advantages Compared with Traditional Storage

Traditional Storage	Cloud Storage
Requires purchase of physical storage hardware	Uses a pay-as-you-use or capacity-based model
Requires data-center space	Provider manages physical facilities
Maintenance and hardware replacement are the Organization's responsibility	Provider manages underlying storage infrastructure
Capacity must often be planned in advance	Capacity can be expanded as needed
Disaster recovery may require a second physical site	Geographic replication and backup services can be added without building another data center

Cloud storage can reduce capital expenditure because the organization does not need to purchase large amounts of hardware before it is needed. It can also reduce operational costs associated with power, cooling, hardware maintenance, and physical storage management. Different storage classes can be selected according to access frequency, with archival storage often being cheaper for long-term backups.

Disaster Recovery Use Case

Suppose a company's primary data center becomes unavailable because of a fire, flood, hardware failure, ransomware incident, or other disaster. If backup copies are stored securely in a separate cloud region, the organization can use those backups to restore databases and application systems.

8. Production data is continuously or periodically backed up to cloud storage.
9. Backup copies are encrypted and access-controlled.
10. Copies are maintained independently of the primary data center.
11. A disaster causes the primary environment to become unavailable.
12. The disaster recovery environment retrieves the required backups.
13. Applications and databases are restored.
14. Business operations resume with less data loss and downtime than would occur without off-site backups.

Conclusion

Cloud storage provides durable, redundant and scalable backup infrastructure. Encryption and access control protect sensitive data, while cloud-based storage can lower infrastructure and maintenance costs. Geographic backup copies are especially valuable for disaster recovery and business continuity.

3. **A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.**

3. CDNs, Edge Servers and Global Cloud Availability

Introduction

A mobile application deployed globally must serve users located far from the application's main cloud region. Network distance, congestion, and server load can increase latency. Content Delivery Networks (CDNs), edge servers, and multi-region cloud deployments help solve these problems.

Content Delivery Network (CDN)

A CDN is a geographically distributed network of servers that delivers content from locations closer to end users. Frequently requested content such as images, videos, JavaScript files, style sheets, application assets, and downloadable files can be cached at edge locations.

How a CDN Reduces Latency

Without a CDN, a user in India might have to request content from a server located in North America or Europe. With a CDN, the request can often be served from an edge location much closer to the user. The shorter network path can reduce latency and improve response time.

Typical flow: User → Nearby CDN Edge Server → Cached Content. If content is not available in the cache, the CDN can retrieve it from the origin and then cache it according to the configured rules.

Benefits of Edge Servers

- Bring frequently accessed content closer to users.

- Reduce round-trip network latency.
- Decrease load on the origin application servers.
- Improve video and media delivery.
- Provide faster delivery of static assets and software packages.
- Improve perceived responsiveness for users in distant regions.

Global Availability Through Cloud Providers
Major cloud providers operate infrastructure in multiple geographic regions and availability zones. Organizations can deploy applications or supporting services in more than one region and use global routing, health checks, replication, backups, and traffic-management mechanisms.

- Multiple regions can serve users in different geographic areas.
- Availability zones provide additional resilience within a region.
- Global DNS or traffic-routing services can direct users toward suitable endpoints.
- Data replication can support continuity between regions.
- CDNs can serve cached content from globally distributed edge locations.

Real-Time Streaming Example
Consider a global video-streaming application. A user in India, another in Europe, and another in the United States should all receive video content without having to contact a single distant origin server for every segment. Popular content can be distributed and cached at CDN edge locations. Users are routed toward suitable nearby locations, reducing latency and improving streaming quality.

For interactive gaming, edge infrastructure can also reduce the distance between players and game services. Lower network latency can improve responsiveness, although real-time game state and server-side processing still require appropriate application architecture.

Conclusion
CDNs and edge servers improve global application performance by bringing content and services closer to users. Multi-region cloud infrastructure adds resilience and availability. Together, these technologies are especially useful for streaming, gaming, social media, and other applications with geographically distributed users.

4. **A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.**

4. Cloud Compliance, Security, Auditing and Data Sovereignty in Financial Institutions

Introduction
Financial institutions handle highly sensitive information, including customer identities, account details, payment records, transaction histories, and financial reports. Cloud deployments therefore require strong

security controls, regulatory compliance, auditing, monitoring, encryption, and data-residency management.

Regulatory Compliance and Auditing

Cloud providers offer security controls, compliance programs, certifications, audit reports, policy tools, identity services, encryption services, logging systems, and other capabilities that can help customers satisfy regulatory requirements. The exact requirements depend on the country, financial regulator, type of data, and services being used.

Compliance is a shared responsibility. The provider secures and operates the underlying cloud infrastructure, while the financial institution remains responsible for correctly configuring its applications, identities, data, permissions, and business processes.

Logging

Cloud logs create an auditable record of events. Examples include login attempts, administrator actions, database access, configuration changes, API calls, file access, and security events. Logs can be retained according to organizational and regulatory requirements.

Monitoring

Monitoring continuously evaluates infrastructure, applications, networks, databases, and security signals. It can detect unusual traffic, repeated failed logins, unexpected configuration changes, abnormal transaction patterns, and service failures. Alerts can be sent to security or operations teams when thresholds or detection rules are triggered.

Encryption

Encryption protects sensitive financial information both at rest and in transit. Databases, storage systems, backups, and other sensitive repositories can use encryption at rest. TLS or equivalent secure protocols can protect information transmitted between customers, applications, APIs, and internal services. Encryption keys must also be protected through appropriate key-management controls.

Data Sovereignty

Data sovereignty means that data is subject to the legal and regulatory requirements of the jurisdiction in which it is stored or processed. A financial organization can address data-residency requirements by selecting approved cloud regions, restricting storage locations through policies, controlling cross-region replication, and auditing where sensitive data is processed.

- Choose cloud regions that satisfy applicable residency requirements.
- Use policy controls to prevent unauthorized geographic movement of data.
- Separate sensitive datasets when different residency rules apply.
- Monitor and audit storage and processing locations.
- Review third-party services to ensure they do not violate data-location requirements.

Sensitive Financial Transaction Example

Consider a banking application that processes an online fund transfer. The customer authenticates using strong identity controls. The mobile application communicates with the cloud service through an encrypted connection. The transaction is validated by application services, stored in an encrypted database, and recorded in audit logs. Monitoring tools look for abnormal activity, while IAM policies restrict access to transaction data.

A simplified flow is: Customer → Secure Mobile App → Encrypted API → Transaction Service → Encrypted Database → Audit Logs and Monitoring.

Conclusion

Cloud providers can provide strong technical and compliance capabilities for financial institutions, but organizations must configure and operate those capabilities correctly. Logging, monitoring, encryption, identity controls, and data-residency policies together provide a strong foundation for protecting financial transactions and meeting regulatory obligations.

5. **A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.**

5. Containerization, Virtual Machines, Kubernetes and Portability

Introduction

Containerization is a software deployment approach in which an application and its required libraries, configuration, and runtime dependencies are packaged into a standardized container image. Containers are widely used for modern microservices and cloud-native applications.

Containers vs Virtual Machines

A virtual machine emulates a complete computer environment and normally includes a guest operating system. A container shares the host operating system kernel while isolating application processes. As a result, containers generally require fewer resources and start faster than full virtual machines.

Feature	Virtual Machine	Container
Operating system	Each VM normally includes a guest OS	Containers share the host OS kernel
Startup time	Usually slower	Usually faster
Resource usage	Higher due to separate OS instances	Lower for many workloads
Image/instance size	Generally larger	Generally smaller
Isolation	Strong hardware/OS-level virtualization isolation	Process and namespace-based isolation
Scaling	Can be slower and heavier	Usually lightweight and fast
Portability	Good	Very high across compatible container platforms

Performance and Resource Usage

Because containers share the host kernel, they avoid the overhead of running a separate full operating system for every application instance. This allows more application workloads to run on the same host in many scenarios. Containers can also start quickly, which is useful for dynamic scaling and continuous deployment.

Virtual machines remain valuable when strong workload isolation, different operating systems, or complete virtualized environments are required. Containers and VMs can also be used together, with containers running inside virtual machines in cloud environments.

Kubernetes Orchestration

Managing a few containers manually may be simple, but a production application may contain hundreds or thousands of containers. Kubernetes is an orchestration platform that automates deployment, scaling, service discovery, health management, and updates for containerized applications.

- Deploys containers according to desired configuration.
- Restarts failed containers automatically.
- Scales workloads up or down.
- Provides service discovery and load balancing.
- Supports rolling updates and controlled rollbacks.
- Maintains the desired number of application replicas.
- Uses health checks to identify unhealthy workloads.

How Kubernetes Simplifies Scaling

Suppose an application normally requires three replicas. When traffic increases, Kubernetes can increase the desired replica count according to configured scaling policies. New containers are scheduled on available nodes. When demand falls, the number of replicas can be reduced. This is much easier than manually starting and stopping individual containers.

Portability Across Cloud Environments

Container images package application dependencies consistently. The same image can be tested on a developer laptop and then deployed to an on-premises Kubernetes cluster or a managed Kubernetes service in a public cloud, provided the required runtime and platform features are supported.

- Reduces environment-specific configuration differences.
- Makes development, testing, staging, and production environments more consistent.
- Supports hybrid-cloud and multi-cloud deployment strategies.
- Simplifies application migration between compatible container platforms.
- Reduces the risk of 'works on my machine' problems.

Microservices-Based E-Commerce Example

An e-commerce system can be divided into independent microservices. Each service can be packaged into its own container and scaled separately.

Microservice	Main Responsibility
User Service	Registration, authentication, profiles, and customer information
Product Service	Product catalog, descriptions, prices, and inventory information
Order Service	Shopping carts and order creation

Payment Service	Payment processing and transaction status
Delivery Service	Shipment and delivery tracking
Notification Service	Email, SMS, or push notifications

Kubernetes can run multiple replicas of these services, restart failed containers, distribute service traffic, and scale high-demand services independently. For example, during a major sale, the Product and Order services may require more replicas than the Notification service.

Conclusion

Containers are lightweight, fast, and portable compared with traditional virtual-machine-based application deployment. Kubernetes provides automated orchestration, scaling, health management, and deployment capabilities. These features make containerization well suited to microservices and modern cloud-native applications.

General Reference Topics

- Cloud computing concepts: elasticity, scalability, availability, redundancy, and pay-as-you-go infrastructure.
- Cloud storage concepts: durability, replication, encryption, access control, backup, and disaster recovery.
- CDN and edge computing concepts: caching, geographic distribution, latency reduction, and global traffic management.
- Cloud security concepts: IAM, encryption, logging, monitoring, compliance, auditing, and data residency.
- Container and Kubernetes concepts: images, containers, orchestration, scaling, service discovery, and rolling updates.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts	Mostly accurate application	Basic application only	Incorrect concept usage

		accurately			
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity